

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Data Structure Practical

Practical – III

Roll No.: F021	Name: LODHI IQRA
Class: SYBsc IT	Subject: DATA STRUCTURE
Date of Assignment: 02/07/2026	Date/Time of Submission: 02/07/2026

Stack Application:
CREATING METHOD OR FUNCTION FOR THIS OPERATION:

```
def push():
    print("push Operation")

def pop():
    print("pop Operation")

def peek():
    print("peek Operation")

def isEmpty():
    print("isEmpty Operation")

num = 0
while (num < 5):
    print("1. Push 2. POP 3. Peek 4. isEmpty 5. Exit")
    num = int(input("Select ch : "))

    if (num == 1):
        push()
    elif (num == 2):
        pop()
    elif (num == 3):
        peek()
    elif (num == 4):
        isEmpty()
    else:
        print("Program Closed")
        print("IQRA S021")
```

```
=====
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 1
push Operation
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 2
pop Operation
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 3
peek Operation
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 4
isEmpty Operation
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 5
Program Closed
IQRA S021
```

Write a program to:

a. Implement a stack using an array.

CODE:

```

import array as arr

stack1 = arr.array('i')

print(stack1)

def push():
    print("push Operation")
    ele = int(input("Enter Element: "))
    stack1.append(ele)
    print(stack1)

def pop():
    print("pop Operation")
    if(isEmpty()==0):
        print("Stack Underflow")
    else:
        stack1.pop()
        print(stack1)

def peek():
    print("peek Operation")
    if(isEmpty()==0):
        print("Stack is Empty")
    else:
        print(stack1[-1])

def isEmpty():
    print("isEmpty Operation")
    if len(stack1)==0:
        return 0

num = 0
while (num<5):
    print("1. Push 2. POP 3. Peek 4. isEmpty 5. Exit")
    num = int(input("Select ch : "))

    if (num==1):
        push()
    elif (num==2):
        pop()
    elif (num==3):
        peek()
    elif (num==4):
        isEmpty()
    else:
        print("Program Closed")
        print("IQRA S021")

```

FOR POSITION

```

import array as arr

stack1 = arr.array('i')

si = int(input("Enter size of stack: "))

print(stack1)

def push():
    print("push Operation")
    if(len(stack1)==si):
        print("Stack Overflow")
    else:
        ele = int(input("Enter Element: "))
        stack1.append(ele)
        print(stack1)

def pop():
    print("pop Operation")
    if(isEmpty()==0):
        print("Stack Underflow")
    else:
        stack1.pop()
        print(stack1)

def peek():
    print("peek Operation")
    if(isEmpty()==0):
        print("Stack is Empty")
    else:
        print(stack1[-1])

def isEmpty():
    print("isEmpty Operation")
    if len(stack1)==0:
        return 0

num = 0
while(num<5):
    print("1. Push 2. POP 3. Peek 4. isEmpty 5. Exit")
    num = int(input("Select ch : "))

    if(num==1):
        push()
    elif(num==2):
        pop()
    elif(num==3):
        peek()

```

```

num = 0
while (num<5):
    print("1. Push 2. POP 3. Peek 4. isEmpty 5. Exit")
    num = int(input("Select ch : "))

    if (num==1):
        push()
    elif (num==2):
        pop()
    elif (num==3):
        peek()
    elif (num==4):
        isEmpty()
    else:
        print("Program Closed")
        print("IQRA S021")

```

OUTPUT:

```

===== RESTART: D:/DATA STRUCTURE SYIT/I
array('i')
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 2
pop Operation
isEmpty Operation
Stack Underflow
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 1
push Operation
Enter Element: 11
array('i', [11])
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 3
peek Operation
isEmpty Operation
11
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 4
isEmpty Operation
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 5
Program Closed
IQRA S021
|

```

```

Enter size of stack: 6
array('i')
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 1
push Operation
Enter Element: 11
array('i', [11])
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 1
push Operation
Enter Element: 12
array('i', [11, 12])
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 1
push Operation
Enter Element: 13
array('i', [11, 12, 13])
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 1
push Operation
Enter Element: 14
array('i', [11, 12, 13, 14])
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 1
push Operation
Enter Element: 15
array('i', [11, 12, 13, 14, 15])
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 1
push Operation
Enter Element: 16
array('i', [11, 12, 13, 14, 15, 16])
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 2
pop Operation
isEmpty Operation
array('i', [11, 12, 13, 14, 15])
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 3
peek Operation
isEmpty Operation
15
1. Push 2. POP 3. Peek 4. isEmpty 5. Exit
Select ch : 4
isEmpty Operation

```